

Fabricio Rodriguez

240-439-5354 | frodrig3@umbc.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Maryland, Baltimore County

Bachelor of Science, Computer Science (GPA: 3.2)

Aug 2022 - May 2026

EXPERIENCE

Financial Industry Regulatory Authority (FINRA)

Jun 2025 - Aug 2025

Software Engineer Intern

Rockville, MD

- Built production automation platform using Python and REST APIs deployed on AWS, autonomously resolving 392 tickets monthly (40% reduction) across 4,000+ devices, directly impacting employee productivity and operational efficiency
- Developed comprehensive CI/CD pipeline using Jenkins and Git with full unit testing coverage, performing phased production rollout from 100 to 4,000 devices while maintaining zero downtime
- Collaborated in an Agile environment with product managers, security stakeholders, and cross-functional teams through daily standups, sprint planning, and code reviews, which improved team alignment and accelerated feature delivery
- Engineered PowerBI analytics dashboards that consumed JSON data from AWS DynamoDB for real-time monitoring, enabling stakeholders to make faster, data-driven decisions
- Won 1st place at FINRA Createathon by rapidly building compliance monitoring POC within 48-hour deadline, presenting technical solution to senior leadership

American University

Jan 2024 - Sep 2024

Machine Learning Undergraduate Researcher

Washington, DC

- Deployed production REST API using Spring Boot (Java) and PostgreSQL supporting 40-50 daily users evaluating \$10+ million in investment opportunities for financial services firm
- Implemented comprehensive unit testing with JUnit and Mockito, maintaining system in active production use 12+ months post-delivery with regular feature updates
- Optimized data processing pipeline achieving 40% performance improvement through algorithmic optimization and vectorized NumPy operations

PROJECTS

Retriever Study - AI Study Matching Platform

Jan 2025 - Present

- Built full-stack production platform serving 150 daily active users using Python FastAPI backend and React TypeScript frontend deployed on AWS (EC2, RDS, S3)
- Architected RESTful API with async processing, OAuth 2.0 authentication, comprehensive error handling, and PostgreSQL database optimization
- Implemented ML matching algorithm using PyTorch achieving 80% accuracy, processing user requests efficiently at scale
- Achieved 150 daily active users through organic adoption, demonstrating product-market fit and user-focused design

RetrieverEats - Campus Delivery Platform

Oct 2025 - Present

- Engineered full-stack application using .NET 8 Web API (C#) and React frontend with PostgreSQL, processing 200+ transactional orders and coordinating workflows for 50+ users
- Implemented Redis caching layer reducing database queries by 40% and optimized SQL queries improving response times by 61% (180ms to 70ms)
- Deployed containerized application using Docker with CI/CD pipeline for automated testing and deployment

TECHNICAL SKILLS

- **Languages:** Java, Python, C#, C++, JavaScript, TypeScript, SQL, HTML/CSS, Go
- **Frameworks:** Spring Boot, .NET 8, React, FastAPI, Flask, Node.js, PyTorch, TensorFlow
- **Developer Tools:** Git, Docker, Jenkins, AWS (EC2, RDS, S3, DynamoDB), PostgreSQL, Redis, JIRA
- **Practices:** Agile/Scrum, CI/CD, Unit Testing (JUnit, pytest, Mockito), REST APIs, OAuth 2.0, Code Reviews